

# Operating system: types, features and examples ★ Golden

One college period · 40–45 minutes · Operating System

## What you will learn

- Define OS and its objectives
- Name Windows, macOS, DOS, UNIX/Linux, Android
- Explain batch, multi-tasking, time-sharing, multi-processing, distributed, embedded
- State that a computer cannot boot without an OS

**Operating System.** System software that manages hardware and software resources and provides an interface between the user and the computer. It is loaded at boot time. A PC cannot boot without an OS.

Common OS: Microsoft Windows, Apple macOS, DOS (old command-line), UNIX and Linux (open source), Android and iOS for mobiles.

| Feature             | Meaning                                        | Example                                |
|---------------------|------------------------------------------------|----------------------------------------|
| Batch processing    | Jobs queued; no user interaction while running | Old payroll runs, mainframes           |
| Multi-tasking       | Several programs appear to run at once         | Browser + Word + music                 |
| Time sharing        | CPU time sliced among users/processes          | University server                      |
| Multi-processing    | Two or more CPUs / cores share work            | Dual-core laptop                       |
| Parallel processing | One job split across processors                | Scientific calculation                 |
| Distributed         | Many computers act as one system               | Google data centres                    |
| Embedded            | OS baked into a device                         | ATM, microwave, car ECU, Android phone |

| CLI (Command Line)           | GUI (Graphical User Interface) |
|------------------------------|--------------------------------|
| Type commands (DIR, CD)      | Windows, icons, mouse          |
| Fast for experts, low memory | Easy for beginners             |
| DOS, UNIX shell              | Windows, GNOME, macOS          |

### HOW BIEK ASKS THIS

Long 2026: "What is an Operating System? Explain its Functions." Functions are the next lecture — keep definition + types here and functions ready.

### COMMON BOARD MISTAKE

Oracle is a database, not an OS. Chrome is a browser (application). Linux is an OS.

آپریٹنگ سسٹم وہ سافٹ ویئر ہے جو کمپیوٹر کو بوٹ کرتا ہے، ہارڈویئر سنبھالتا ہے اور صارف کو انٹرفیس دیتا ہے۔ ونڈوز، لینکس، اینڈرائیڈ مشہور OS ہیں۔  
ملٹی ٹاسکنگ کئی پروگرام ایک ساتھ؛ ایمبیڈڈ OS فریج یا موبائل کے اندر ہوتا ہے۔

#### CHECK YOURSELF

1. Can a computer boot without an OS?
2. Give one embedded OS example.

Answers: 1. No · 2. Android in a phone / OS in an ATM

#### MCQs

1. A computer cannot boot if it does not have a:
  - a. Compiler
  - b. Browser
  - c. Operating System
  - d. Loader only
2. Which is NOT a type of operating system?
  - a. Windows
  - b. Linux
  - c. Mac OS
  - d. Oracle
3. Windows is:
  - a. Application software
  - b. System software
  - c. A compiler
  - d. Firmware only

Key: 1=c, 2=d, 3=b

#### Short questions (3 marks each)

1. Define operating system. Name four operating systems.
2. Differentiate multi-tasking and multi-processing.
3. What is an embedded operating system?

#### Model short answers

1. OS is system software that manages resources and lets the user run programs. Windows, Linux, macOS, Android.
2. Multi-tasking: many programs on one CPU by time-slicing. Multi-processing: more than one processor executing at the same time.
3. An OS built into a dedicated device (router, car, TV, mobile) with a specific job.

#### Long questions (board Section C)

1. What is an operating system? Explain its types / features with examples.